

Performance Testing

Date	3 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate the responsiveness, stability, and reliability of the SmartLender web application during loan eligibility prediction. The testing verified that the application performs efficiently under normal user load and provides prediction results without significant delay.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Web Browser (Chrome), Flask Local Server
Type of Testing	Load Testing, Response Time Testing
Target Module / API	Loan Eligibility Prediction Module (/predict)
Test Environment	Local System & Hugging Face Deployment
Test Date	3 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid applicant details	10	60	Loan prediction generated successfully
2	Multiple prediction requests	20	60	Application remains responsive
3	Invalid input submission	10	30	Input validation message displayed
4	Continuous prediction requests	25	120	Stable performance without errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.8 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)		18 Req/sec	Pass	Stable request handling
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	52%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	58%	Pass	Memory usage within limit

Step 4: Observations & Analysis

Key Findings

- The application processed loan prediction requests efficiently.
- Prediction results were generated quickly with low response time.
- No significant performance degradation was observed during testing.
- The Flask application remained stable under multiple requests.

Bottlenecks Identified

- Minor delay observed when the application was started for the first time due to model loading.
- No critical bottlenecks were identified.

Optimization Steps Taken

- Optimized data preprocessing before prediction.
- Loaded the trained XGBoost model once during application startup.
- Reduced unnecessary processing to improve response time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6

output):

SmartLender

ML-Powered Credit Evaluation

AI-POWERED

Smart Loan Approval Prediction System

Instantly evaluate loan applications using advanced machine learning algorithms. Our XGBoost model analyses applicant profiles to deliver fast, accurate, and unbiased credit decisions.

Start Prediction

Applicant Details Form

Fill in all fields below to get an instant loan approval prediction

PERSONAL INFORMATION

Gender

Male

Marital Status

Married

Number of Dependents

1

Education Level

Graduate

Enter the number of people financially dependent on you

EMPLOYMENT & INCOME

Self Employed

Yes

Applicant Monthly Income (₹)

70000

Co-Applicant Monthly Income (₹)

SmartLender

Prediction Result



LOAN PREDICTION RESULT

Loan will be Approved ✓

Based on the applicant's profile, our XGBoost model has determined that this loan application meets the credit criteria. The applicant demonstrates a positive credit standing and sufficient repayment capacity.

New Prediction

Home